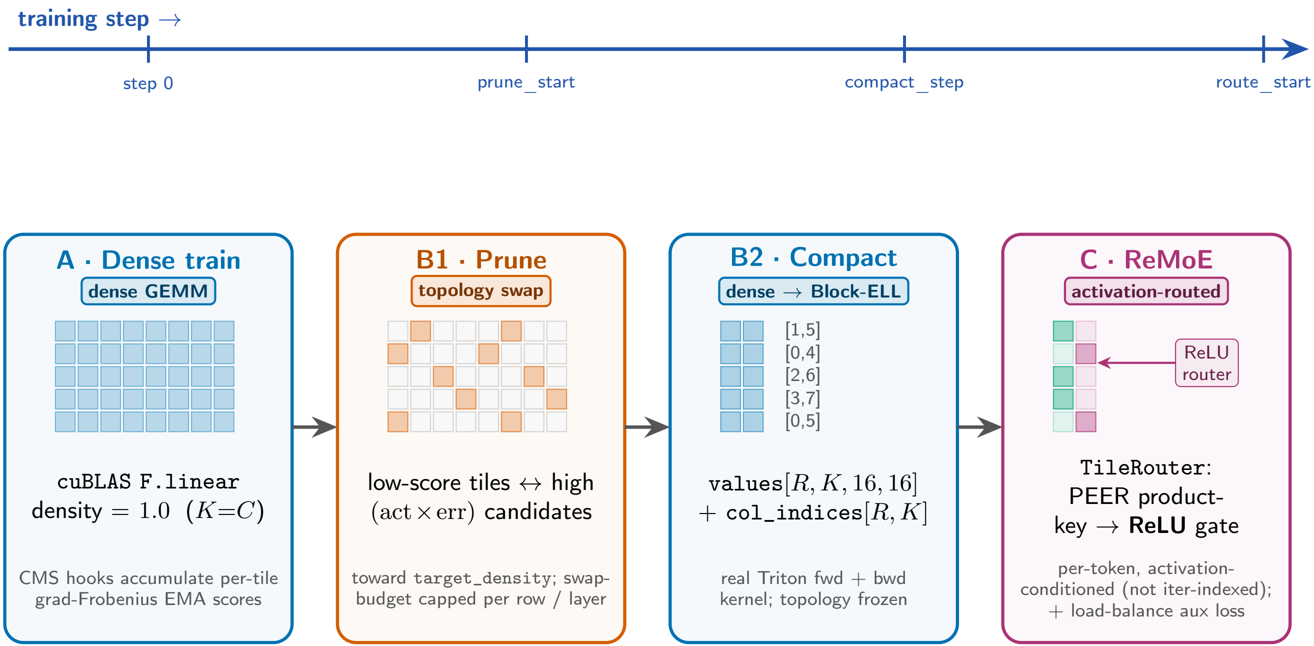
MORPH — Core-MLP Lifecycle: Block-ELL · CMS Pruning · ReMoE



Block-ELL format: 16×16 tiles; $R = d_{\text{out}}/16$ rows, $C = d_{\text{in}}/16$ cols, K active block-cols per row, **density** $= K/C$.
Scope: all SwiGLU MLPs (`gate_up` + `down`) across **prelude** + **core** + **coda** (`block_ell_scope=all`); the core block is re-applied every loop iteration.

CMS schedule: accumulate scores every step · `score_step` every 10 · topology swaps periodically from `prune_start` · `compact()` once at `compact_step` · ReMoE at `route_start`. Tile scoring = gradient-Frobenius EMA (incumbents) vs. $\mathrm{err}[r] \times \mathrm{act}[c]$ (candidates).

ReMoE: routing is *per-token activation-conditioned* on each Block-ELL layer (prelude / core / coda). In the looped core, different iterations route differently only because their activations differ — *not* via any iteration-index input.